

Soln: $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\therefore$ characteristic eqn

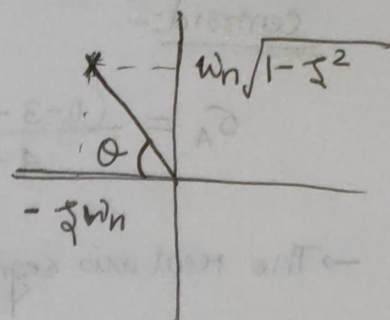
$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$\cos\theta = \frac{+\zeta\omega_n}{\sqrt{\zeta^2\omega_n^2 + \omega_n^2 - \omega_n^2\zeta^2}}$$

$$\cos\theta = \frac{+\zeta\omega_n}{\omega_n}$$

$$\Rightarrow \theta = \cos^{-1}(\zeta)$$



Ex: A feedback control system has CLTF -

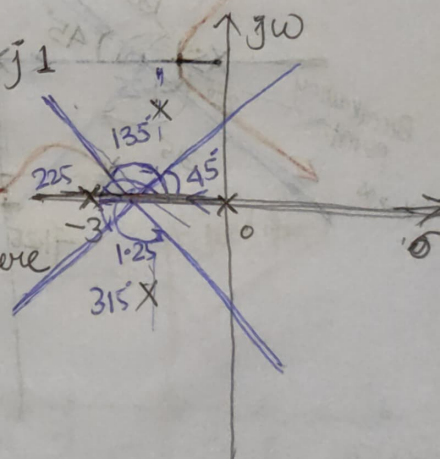
$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Root locus as K vary from zero to infinity.

Soln: characteristic eqn:

The open loop poles are: $0, -3, -1 \pm j1$

All four branches will go to infinity along straight line asymptotes, since there are no open loop zeros.



Asymptote angles are given by -

$$\phi_A = \frac{(2q+1)180^\circ}{n-m}, q=0,1,2,3$$

$$= \frac{(2q+1)180^\circ}{4-0}, q=0,1,2,3$$

$$= 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

Centroid:-

$$\sigma_A = \frac{(0-3-1-j-1)}{4-0} = -1.25$$

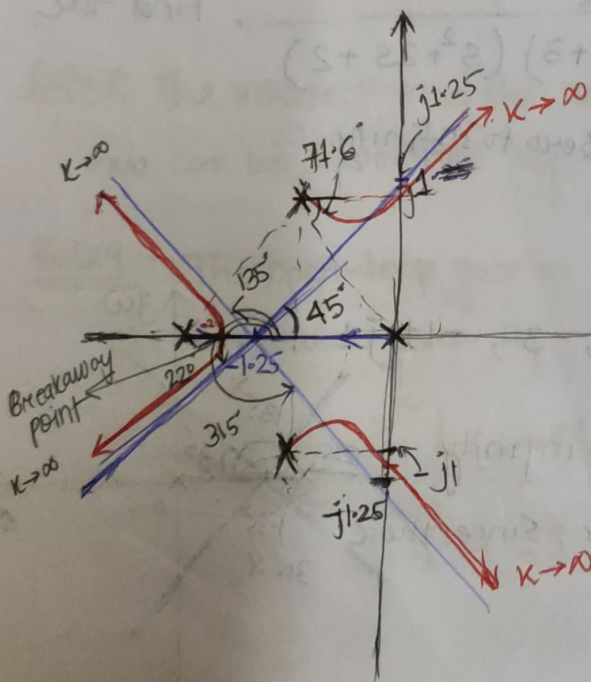
→ The real axis segment 0 to -3 is a part of the root locus.

Characteristic eqn;

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\therefore K = -s(s+3)(s^2+2s+2)$$

$$\therefore \frac{dK}{ds} = 0 \Rightarrow s = -2.3, -0.725 \pm j0.365$$



The angle of departure from ~~open-loop~~ upper complex pole $(-1 \pm j1)$

$$\begin{aligned} \phi_p &\equiv \pm 180^\circ (2q+1) + \phi \\ &\equiv \pm 180^\circ + (-135^\circ - 90^\circ - 26.6^\circ) \\ &= -71.6^\circ \end{aligned}$$

Characteristic eqn:

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\therefore s^4 + 5s^3 + 8s^2 + 6s + K = 0$$

The Routh array is -

$$\begin{array}{ccc|c} s^4 & 1 & 8 & K \\ s^3 & 5 & 6 & \\ s^2 & \frac{34}{5} & K & \\ s^1 & \frac{204}{5} - 5K & 0 & \\ s^0 & \frac{34}{5} & & \\ s^0 & K & & \end{array}$$

when,

$$\frac{\frac{204}{5} - 5K}{\frac{34}{5}} = 0$$

$$\Rightarrow K = \frac{204}{25}$$

$\therefore$ The auxiliary eqn is -

$$\frac{34}{5} s^2 + \frac{204}{25} = 0$$

$$\therefore s = \pm j1.1$$

- Q. Find the roots corresponding to $\zeta = 0.7 \rightarrow$ constant $\zeta = 0.7$ line with angle $\cos^{-1}(\zeta)$
- Q. Find 'K' at breakaway point

Characteristic eqn.

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\Rightarrow -K = |s| |s+3| |s^2+2s+2| \Big|_{s=-2.3}$$

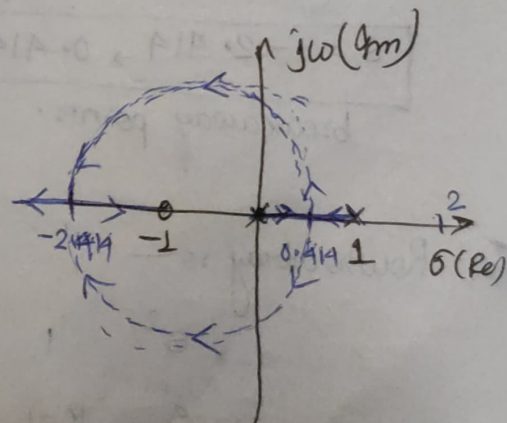
$$\Rightarrow K = 4.54$$

* Control Systems (Tutorials):

Root Locus:-

eg 1: $G(s) = \frac{K(s+1)}{s(s-1)}$

soln: O/L poles $\neq s=0, 1$
O/L zeros, $s=-1$



The points betw 0 to 1 is a part of root-locus because the no. of poles/zeros to the right of it is odd.

Asymptote angle, $\phi_A = \frac{(2q+1) 180^\circ}{n-m}$, $q=0, 1, \dots, n-m-1$

$2-1-1=0$

$q=0$

$$= \frac{180^\circ}{2-1}$$

$$= 180^\circ$$

Centroid = $\frac{0+1-(-1)}{2-1} = 2$

→ Characteristic eqn is—

$$1 + \frac{K(s+1)}{s(s-1)} = 0 \Rightarrow K = \frac{-s(s-1)}{s+1}$$

$$\Rightarrow K(s+1) + s(s-1) = 0 \quad \therefore \frac{dK}{ds} = 0$$

$$\Rightarrow s^2 + (K-1)s + K = 0 \Rightarrow \left[\frac{(s+1)[s-1] + s - s(s-1)}{(s+1)^2} \right] = 0$$

$$\Rightarrow K =$$

$$\therefore s_{1,2} = \frac{-2 \pm \sqrt{4+4}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$= -1 \pm \sqrt{2} = -1 - 1.414$$

$-1 + 1.414$

$$s = -2.414, 0.414$$

breakaway points.

$$\Rightarrow (s+1)(s-1) + s(s+1) - s(s-1) = 0$$

$$\Rightarrow (s^2-1) + s(s+1-s+1) = 0$$

$$\Rightarrow s^2-1 + 2s = 0$$

$$\Rightarrow s^2 + 2s - 1 = 0$$

Routh array is—

s^2	1	K
s^1	K-1	.
s^0	K	

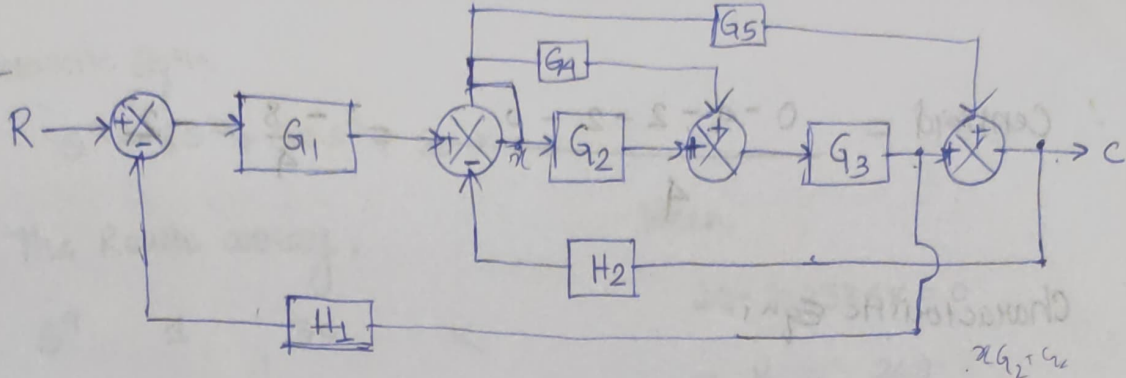
When $K=1$

Auxiliary polynomial eqn

$$s^2 + 1 = 0$$

$$\Rightarrow s = \pm j$$

Eg:-



Q. Draw the root-locus for a system, The open-loop T.f is given

by $G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$, show all the ~~stability~~ ~~ending~~ points on the root locus.

Soln:-

$$G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$

No. of poles = 4

Open-loop poles, $s = 0, -4, -2 \pm j4$

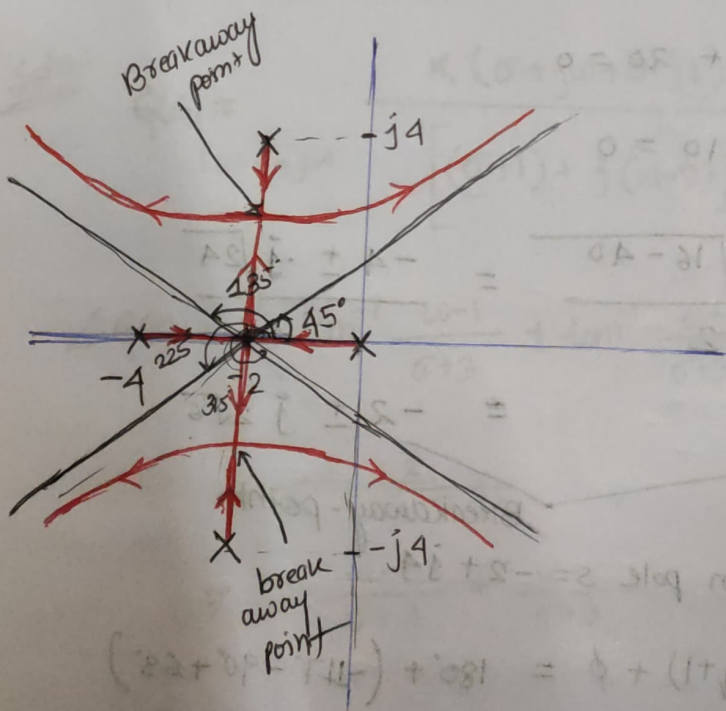
$$s^2 + 4s + 20 = 0$$

$$\Rightarrow s = \frac{-4 \pm \sqrt{16 - 80}}{2}$$

$$= \frac{-4 \pm \sqrt{-64}}{2}$$

$$= \frac{-4 \pm j8}{2}$$

$$= -2 \pm j4$$



Asymptote angle, $\phi_A = \frac{(2q+1)180^\circ}{n-m}$, $q = 0, 1, 2, \dots, n-m-1$

$$= \frac{(2q+1)180^\circ}{4} = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

$$\therefore \text{Centroid} = \frac{0 - 4 - 2 - 2 - 0}{4} = -\frac{8}{4} = -2$$

Characteristic Eqn is -

$$1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{S(S+4)(S^2+4S+20)} = 0$$

$$\Rightarrow K = -S(S+4)(S^2+4S+20) = -(S^2+4S)(S^2+4S+20)$$

$$\therefore \frac{dK}{ds} = 0$$

$$\Rightarrow -[(S^2+4S)(2S+4) + (2S+4)(S^2+4S+20)] = 0$$

$$\Rightarrow (2S+4)(S^2+4S + S^2+4S+20) = 0$$

$$\Rightarrow 2S^2 + 8S + 20 = 0$$

$$\Rightarrow S^2 + 4S + 10 = 0$$

$$S = \frac{-4 \pm \sqrt{16 - 40}}{2} = \frac{-4 \pm j\sqrt{24}}{2}$$

$$\therefore S = -2 \pm j\sqrt{6}$$

Breakaway-point

$\therefore$ Angle of departure from pole $s = -2 + j4$

$$\theta_p = 180^\circ (2q+1) + \phi = 180^\circ + (-117^\circ - 90^\circ + 63^\circ)$$

$$= -90^\circ$$

Characteristic Eqn;

$$s^4 + 8s^3 + 36s^2 + 80K = 0$$

∴ The Routh array,

$$\begin{array}{ccc} s^4 & 1 & 36 & K \\ s^3 & 8 & 80 & 0 \\ s^2 & 26 & K & \\ s^1 & \frac{260-K}{26} & 10-0.0384K & \\ s^0 & K & & \end{array}$$

When,

$$10 - 0.0384K = 0$$

$$\Rightarrow K = 260$$

∴ The auxiliary polynomial is

$$26s^2 + K = 0$$

$$\Rightarrow 26s^2 + 260 = 0$$

$$\Rightarrow s^2 = -10$$

$$\Rightarrow s = \pm j\sqrt{10}$$

Q. Show that the root locus plot for a control system with

$$G(s) = \frac{K(s^2 + 6s + 10)}{s^2 + 2s + 10}, \quad H(s) = 1 \quad \text{is a circle centered at origin with radius } \sqrt{10}.$$

Soln:-

$$G(s) = \frac{K(\sigma + j\omega + 3 - j1)(\sigma + j\omega + 3 + j1)}{[(\sigma + 1) + j(\omega - 3)][(\sigma + 1) + j(\omega + 3)]}$$

$$\angle G(s) = \left(-\tan^{-1} \frac{\omega - 1}{\sigma + 3} + \tan^{-1} \frac{\omega + 1}{\sigma + 3} \right) - \left(\tan^{-1} \frac{\omega - 3}{\sigma + 1} + \tan^{-1} \frac{\omega + 3}{\sigma + 1} \right) = 180^\circ$$

$$\Rightarrow \sigma^2 + \omega^2 = 10$$

$$\Rightarrow \frac{\omega - 1}{\sigma + 3} + \frac{\omega + 1}{\sigma + 3}$$

Q. The open-loop Transfer function of a system is given by -

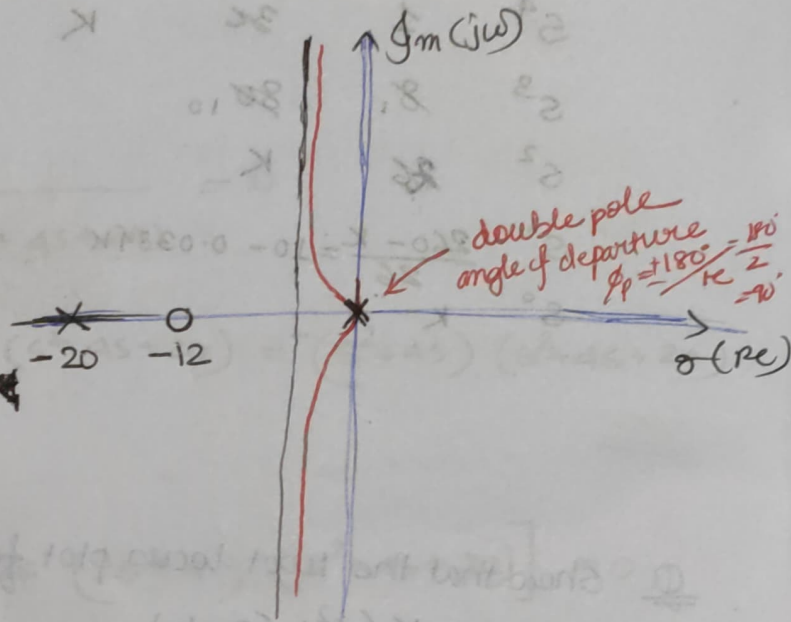
$$G(s) H(s) = \frac{K (s+12)}{s^2 (s+20)}$$

Sketch the root-locus plot.

Soln: poles, $s = 0, 0, -20$

Zeros, $s = -12$

$$\begin{aligned} \text{Centroid} &= \frac{0+0-20-(-12)}{3-1} \\ &= \frac{-20+12}{2} = \frac{-8}{2} = -4 \end{aligned}$$



Q. Investigate the stability: $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$

Soln:

s^5	1	2	3		$\frac{5s+2}{s}$
s^4	1	2	5		$\frac{2s+2}{s}$
s^3	$\frac{0(+)}{(-)}$	-2			$\frac{5s^2+4s+4}{2s+2}$
s^2	$\frac{2s+2}{s}$	5			$\frac{5s^2+4s+4}{2s+2}$
s^1	-	$\frac{5s^2+4s+4}{2s+2}$			
s^0	5(+)				

There are two sign change $(+) \rightarrow (-) \rightarrow (+)$

$\therefore$ System is unstable with two poles on Right half of s-plane.

Q. $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$

$$\begin{array}{rcl} s^6 & 1 & 8 \quad 20 \quad 16 \\ s^5 & 2 & 12 \quad 16 \quad 8 \\ s^4 & -2 & -12 \quad -16 \\ s^3 & 0(E) & 0(E) \\ s^2 & 0(E) & \\ s^1 & & \\ s^0 & & \end{array}$$

$\therefore$ The auxiliary polynomial is -

$$A(s) = -2s^4 - 12s^2 - 16 = 0$$

$$\begin{array}{rcl} s^4 & -2 & -16 \\ s^2 & -12 & 0 \\ s^0 & & \end{array}$$

$$\frac{dA(s)}{ds} = -8s^3 - 24s = 0$$

$$A(s) = s^4 + 6s^2 + 8 = 0$$

$$\begin{array}{rcl} s^6 & 1 & 8 \quad 20 \quad 16 \\ s^5 & 2 & 12 \quad 16 \quad 8 \\ s^4 & +2 & +12 \quad +16 \quad 8 \\ s^3 & +8 & +24 \quad 3 \\ s^2 & 3 & 8 \\ s^1 & \frac{1}{3} & 0 \\ s^0 & 8 & \end{array}$$

$$\therefore s = \pm j\sqrt{2}$$

$$\text{and } s = \pm j2$$

$\Rightarrow$ marginally stable.

Q. The open-loop Transfer function of a unit feedback control system is given - $G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$, by applying

Routh criteria. Discuss the stability of the closed loop system as a function of K . Determine the value of K which will cause sustained oscillation in the CL system.

What are the corresponding oscillation frequencies.

Soln

$$C:L \text{ TF.} = \frac{1}{1 + G(s)H(s)} = \frac{1}{1 + \frac{K}{(s+2)(s+4)(s^2+6s+25)}}$$

$\therefore$ The

∴ The characteristic Eqn -

$$1 + \frac{K}{(s+2)(s+4)(s^2+6s+25)} = 0$$

$$\Rightarrow K = -(s+2)(s+4)(s^2+6s+25)$$

$$\Rightarrow s^4 + 12s^3 + 69s^2 + 198s + (200+K) = 0$$

The Routh array is -

s^4	1	69	$200+K$
s^3	12	198	
s^2	$\frac{52.5}{43.5}$	$200+K$	
s^1	$198 - \frac{12(200+K)}{52.5}$		
s^0	$200+K$		

For s-stability

$$198 - \frac{12(200+K)}{52.5} > 0$$

$$\Rightarrow \text{and } 200+K > 0$$

$$\Rightarrow K > -200$$

$$\frac{12(200+K)}{52.5} < 198$$

$$\Rightarrow K < 666.25$$

∴ Range of K:

$$-200 < K < 666.25$$

For sustained oscillation,

$$198 - \frac{12(200+K)}{52.5} = 0$$

$$\Rightarrow K = 666.25$$

∴ The auxiliary Eqn -

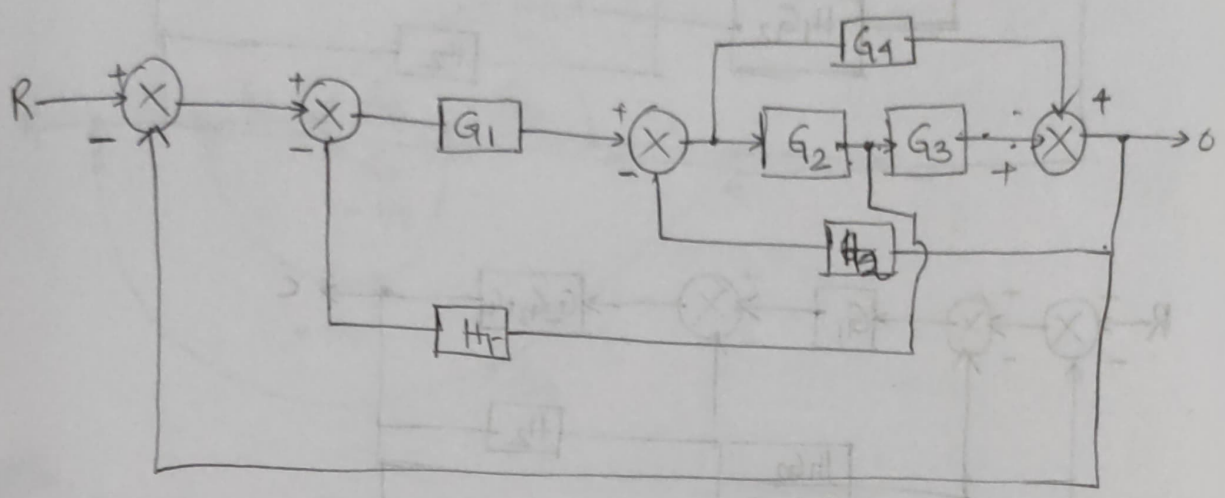
$$52.5s^2 + 200 + 666.25 = 0$$

$$\Rightarrow s = \pm j 4.06$$

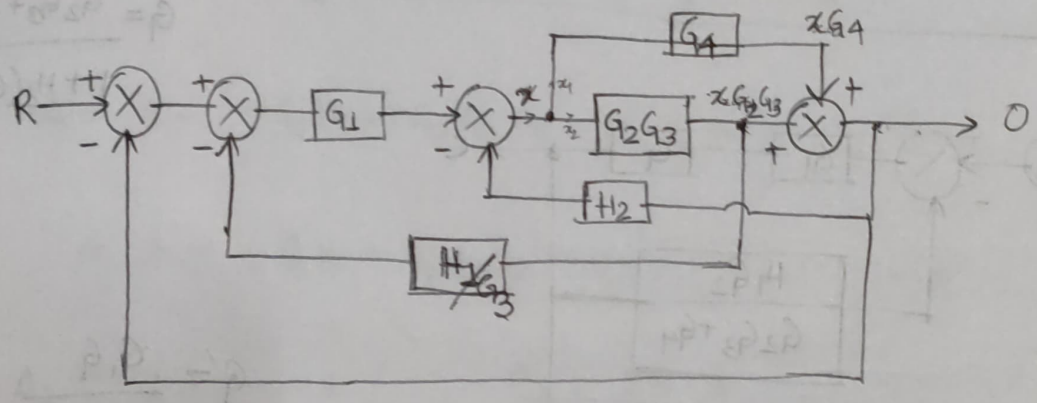
∴ frequency of oscillation = 4.06 rad/s

Assignment:-

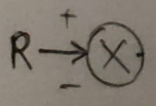
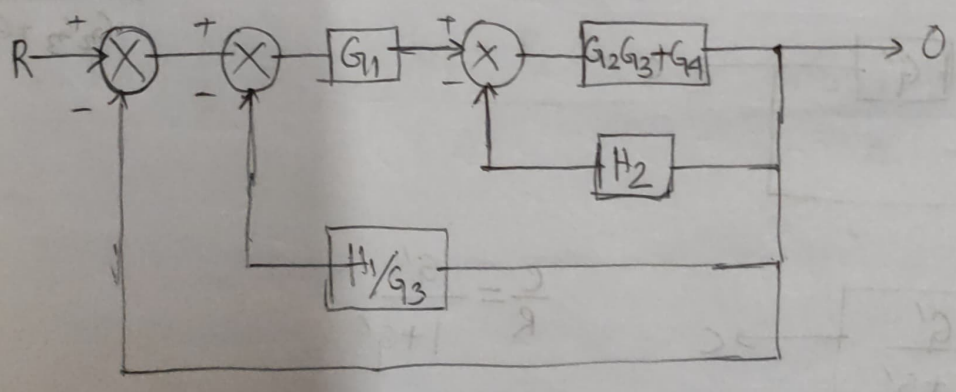
- Q1: Obtain the block diagram and Tf of armature control DC motor.
- Q2: Obtain the block diagram and Tf of field control DC motor.

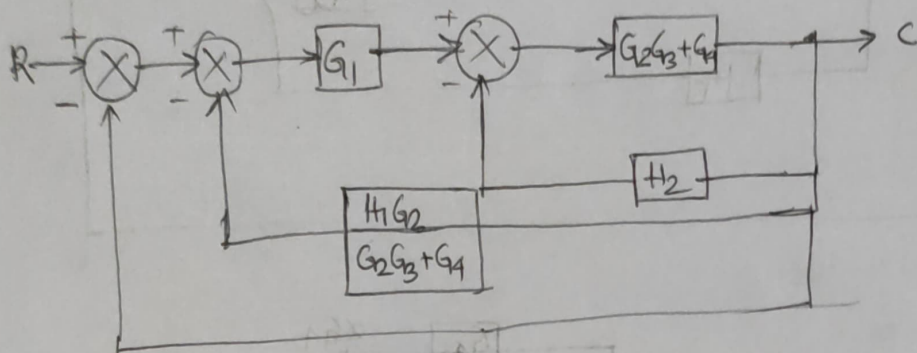
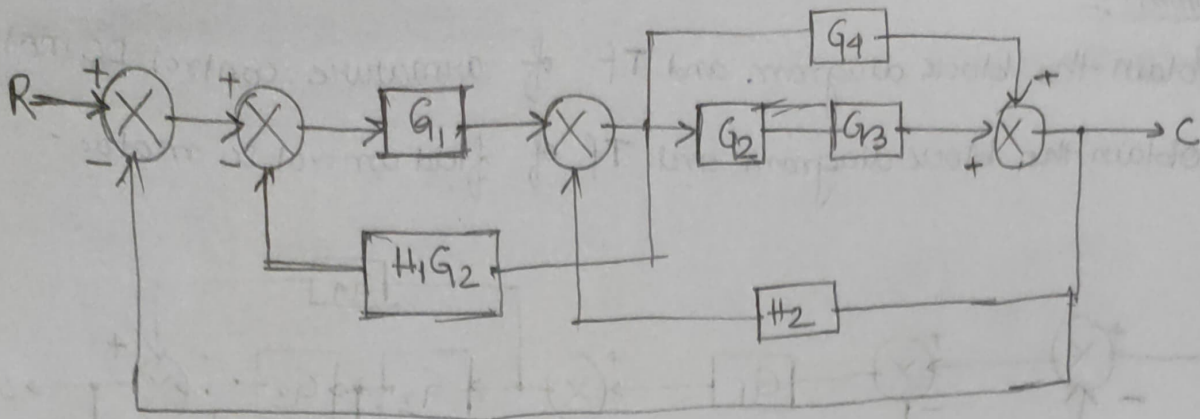


Soln:

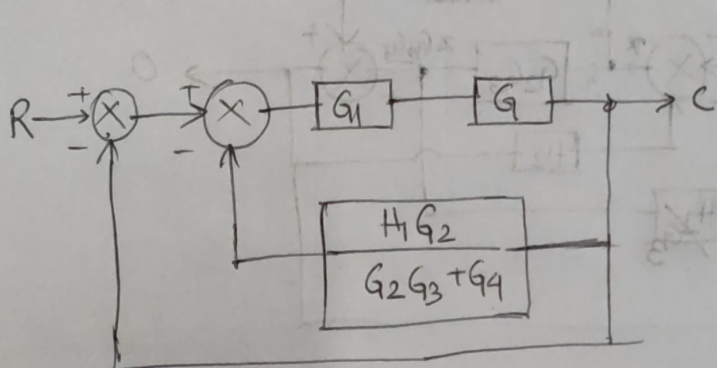


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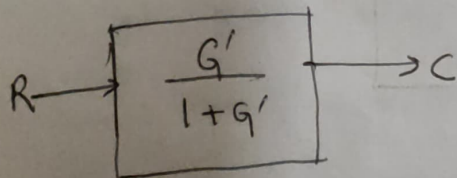
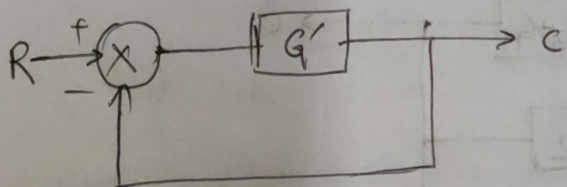




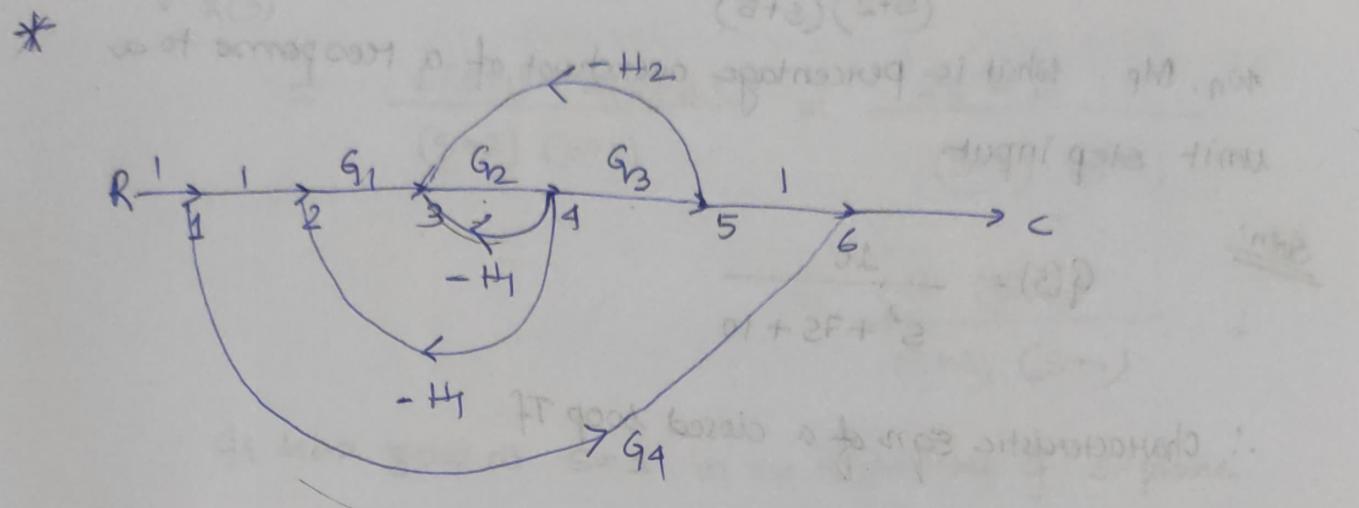
$$G = G_2G_3 + G_4$$



$$G' = \frac{G_1G}{1 + H_1G_2G_1G + G_2G_3 + G_4}$$



$$\frac{C}{R} = \frac{G'}{1 + G'}$$



$$\Delta = 1 - (-G_2 H_1 - G_2 G_3 H_2 - G_1 G_2 H_1)$$

$$\Delta_1 = 1 - 0 = 1, \quad \Delta_2 = \underline{1} - \Delta$$

$$\therefore T = \frac{C}{R} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 \Delta}{\Delta}$$

Q. The open loop Tf of a system is given by -

$$G(s) = \frac{10}{(s+2)(s+5)}$$

ω_n , M_p . What is percentage overshoot of a response to a unit step input.

Soln:

$$G(s) = \frac{10}{s^2 + 7s + 10}$$

$\therefore$ Characteristic eqn of a closed loop Tf.

$$1 + G(s)H(s) = 0$$

$$\Rightarrow \frac{10}{s^2 + 7s + 10} + 1 = 0$$

$$\Rightarrow s^2 + 7s + 20 = 0$$

$$\therefore 2\zeta\omega_n = 7, \omega_n^2 = 20$$

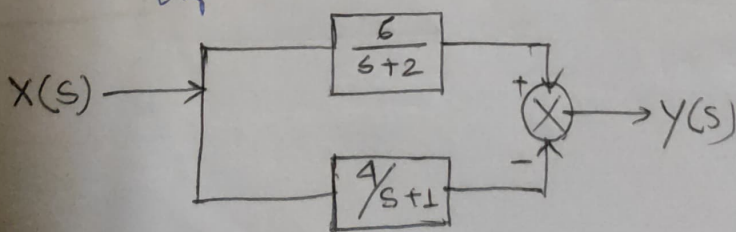
$$\Rightarrow \zeta = \frac{7}{2 \times 2\sqrt{5}}$$

$$= 0.7826$$

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = 1.92\%$$

$$\Rightarrow \omega_n = \pm\sqrt{20} = \pm 2\sqrt{5} = 4.472 \text{ rad/sec}$$

Q. Show that a system Tf $\frac{Y(s)}{X(s)}$ has a zero in the right half plane. obtain $y(t)$. $x(t)$ is a unit step function for the system. shown in fig.



Soln:

$$\begin{aligned}\frac{Y(s)}{X(s)} &= \frac{6}{s+2} - \frac{4}{s+1} \\&= \frac{6(s+1) - 4(s+2)}{(s+2)(s+1)} = \frac{2s-2}{(s+2)(s+1)} \\&= \frac{2(s-1)}{(s+2)(s+1)}\end{aligned}$$

$\therefore$ It has zero at $s=1$, in the right half of s -plane.

$$\begin{aligned}\Rightarrow Y(s) &= \frac{2(s-1)}{(s+2)(s+1)} \cdot \frac{1}{s} \\&= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}\end{aligned}$$

$$\therefore A = \left. \frac{2(s-1)}{(s+2)(s+1)} \right|_{s=0} = \frac{-2}{2} = -1$$

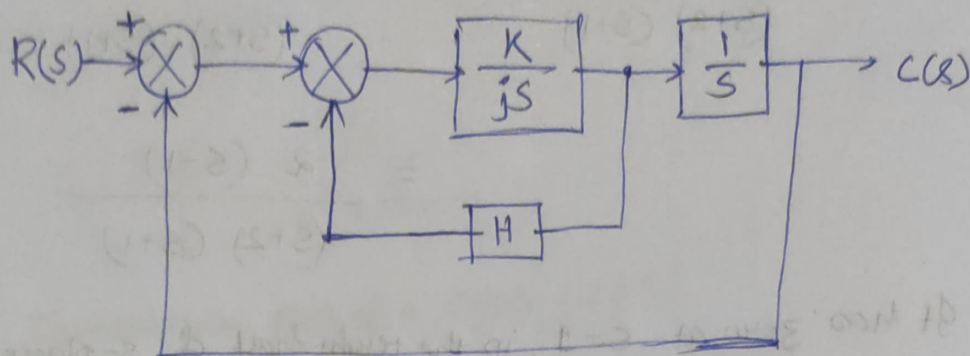
$$B = \left. \frac{2(s-1)}{(s+2)s} \right|_{s=-1} = \frac{2(-1-1)}{(-1+2)(-1)} = \frac{-4}{+1} = -4$$

$$C = \left. \frac{2(s-1)}{(s+1)s} \right|_{s=-2} = \frac{2(-2-1)}{(-2+1)(-2)} = \frac{-6}{2} = -3$$

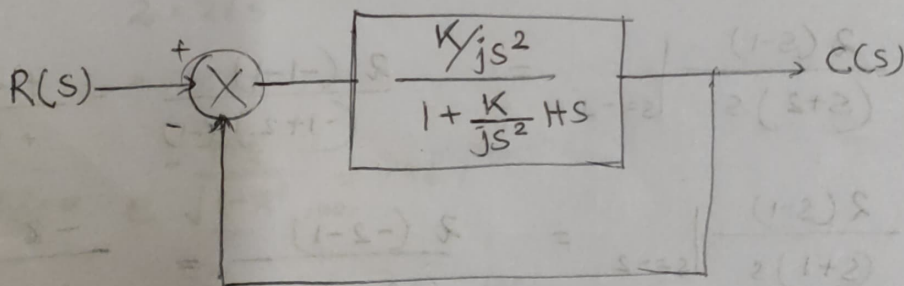
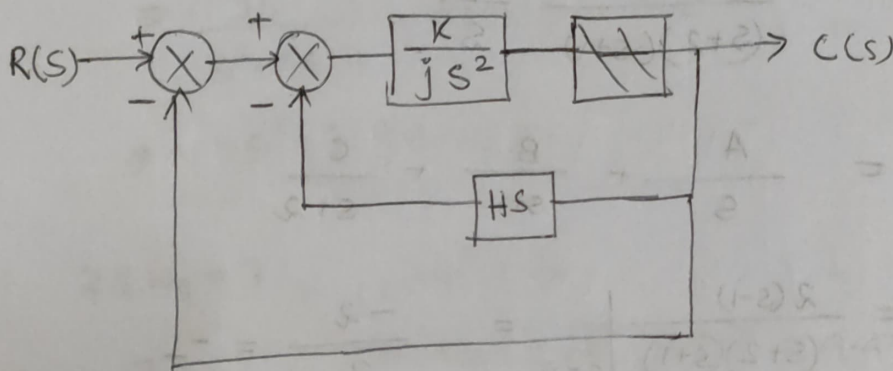
$$\therefore Y(s) = \frac{-1}{s} - \frac{4}{s+1} - \frac{3}{s+2}$$

$$\begin{aligned}\therefore y(t) &= -u(t) + 4e^{-t}u(t) - 3e^{-2t}u(t) \\&= (4e^{-t} - 3e^{-2t} - 1) \quad t \geq 0\end{aligned}$$

Q. Determine the value of K and H of a closed loop system so that max^m overshoot of a unit step response is 25% and peak time $t_p = 2 \text{ sec}$. Assume $J = 1 \text{ kg m}^2$



Solnⁿ



$$\frac{C(s)}{R(s)} = \frac{\frac{K}{Js^2}}{1 + \frac{KH}{Js}} = \frac{K}{Js^2 + KH}$$

$$= \frac{K}{Js^2 + \frac{KH}{J}}$$

$$\frac{CCS}{RCS} = \frac{K/s}{s^2 + \frac{KH}{J}s + \frac{K}{J}}$$

$$\therefore 2\zeta \omega_n = KH$$

$$\omega_n^2 = \frac{K}{J}$$

$$\Rightarrow \zeta = \frac{KH}{2\sqrt{K}}$$

$$\Rightarrow \omega_n = \pm \sqrt{\frac{K}{J}}$$

$$\therefore M_p = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} = 0.25$$

$$= e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} = \ln 0.25$$

$$\Rightarrow \frac{\zeta}{\sqrt{1-\zeta^2}} = 0.44127$$

$$\therefore t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\sqrt{K} \sqrt{1-\zeta^2}} = 2$$

$$\Rightarrow \frac{1}{\sqrt{K} \sqrt{1-\zeta^2}} = 0.6326$$

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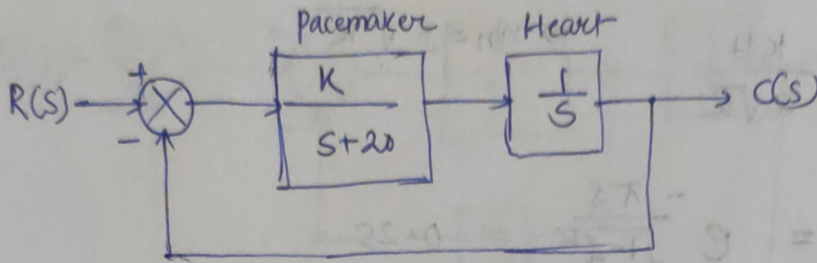
$$\therefore \frac{①}{②} \Rightarrow \frac{\zeta}{1/\sqrt{K}} = \frac{0.44127}{0.6326}$$

$$\Rightarrow \zeta \sqrt{K} = 0.69316$$

$$\Rightarrow \frac{KH}{2} = 0.69316$$

$$\Rightarrow KH = 1.3863$$

Q. Block diagram of an electronic pacemaker given in determine the steady state error of unit ramp when $K=400$ also determine the value of K for which the steady state error to unit ramp is 0.02 .



Soln:

$$\frac{C(s)}{R(s)} = \frac{\frac{K}{s+20}}{1 + \frac{K}{(s+20)s}}$$

$$R(s) = \frac{1}{s^2}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + \frac{K}{(s+20)s}}$$

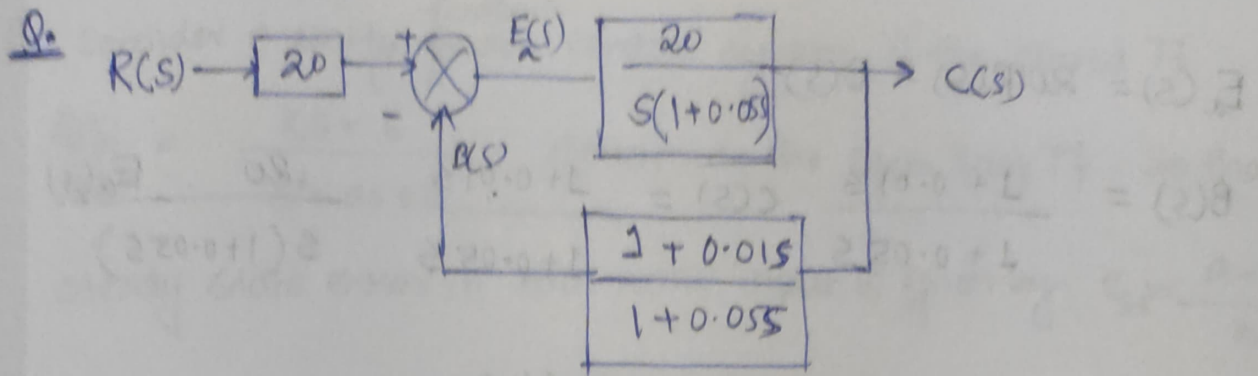
$$= \lim_{s \rightarrow 0} \frac{s^2 (s+20) R(s)}{(s+20)s + K}$$

$$= \lim_{s \rightarrow 0} \frac{s+20}{s^2 + 20s + K}$$

$$\Rightarrow \frac{20}{K} = 0.02$$

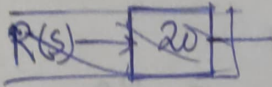
$$\Rightarrow K = \frac{20}{0.02} = 1000$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s+20}{s^2 + 20s + 400} = \frac{20}{400} = 0.05$$



For a given system shown in fig. Determine the actuating signal $E_a(s)$. Also determine the position error constant K_p for unit step input.

Soln:



$$E_a(s) = 20R(s) - \frac{1+0.01s}{1+0.05s} C(s)$$

$$= \frac{20}{s} - \frac{1+0.01s}{1+0.05s} C(s)$$

$$E_a(s) = \frac{20(1+0.05s) - (1+0.01)s C(s)}{s(1+0.05s)}$$

$$C(s) = \frac{\frac{20}{s(1+0.05s)}}{1 + \frac{20}{s(1+0.05s)} \times \frac{1+0.01s}{1+0.05s}} \times 20R(s)$$

$$= \frac{400}{s(1+0.05s)^2} R(s)$$

$$E_a(s) = 20 R(s) - B(s)$$

$$B(s) = \frac{1+0.01s}{1+0.05s} CCS = \frac{1+0.01s}{1+0.05s} \cdot \frac{20}{s(1+0.05s)} E_a(s)$$

$$\therefore E_a(s) = 20 R(s) - \frac{(1+0.01s) 20}{s(1+0.05s)^2} E_a(s)$$

$$\Rightarrow E_a(s) \left[1 + \frac{(1+0.01s) 20}{s(1+0.05s)^2} \right] = 20 R(s)$$

$$\Rightarrow E_a(s) \left[\frac{s(1+0.05s)^2 + (1+0.01s) 20}{s(1+0.05s)^2} \right] = \frac{20}{s}$$

$$\Rightarrow E_a(s) = \frac{20(1+0.05s)^2}{s(1+0.05s)^2 + 20(1+0.01s)}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E_a(s) = \lim_{s \rightarrow 0} \frac{s 20 (1+0.05s)^2}{s(1+0.05s)^2 + 20(1+0.01s)}$$

$$= \frac{20}{20} = 1$$

$$\therefore \frac{1}{1+K_p} = 0 \Rightarrow K_p = \infty$$

Q. Consider a unity ~~factor~~^{feedback} control system, if the closed Tf

$$\frac{G(s)}{R(s)} = \frac{ks + 6}{s^2 + as + 6}$$

Determine the open loop Tf. So that

steady state error in unit ramp input is given by $e_{ss} = \frac{a-k}{b}$